Generative Deep Learning Pipeline for the Rapid Design of Patient-Specific Therapeutic Binders Against Customized Cancer Markers

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Background: The MMP Selectivity Problem

Matrix Metalloproteinases (MMPs) are zinc-dependent endopeptidases essential to extracellular matrix remodeling [1]. In cancer, MMP9 is the primary driver of basement membrane degradation, enabling tumor cell intravasation and metastatic spread; it additionally promotes angiogenesis by cleaving and releasing VEGF sequestered in the ECM [3]. The clinically relevant distinction: MMP2 maintains basal connective tissue homeostasis, while MMP9 is the pathogenic species in >90% of invasive breast, colorectal, and lung cancers. Despite this, MMP2 and MMP9 share near-identical active site architecture (~65% identity in the catalytic domain), making selective inhibition extraordinarily difficult. Previous broad-spectrum MMP inhibitors (marimastat, batimastat) failed Phase III oncology trials due to musculoskeletal toxicity caused by off-target inhibition of MMP1/2/3 [3]. TIMP3 is a natural, endogenous inhibitor that engages metalloproteinase active sites through its N-terminal reactive site loops—offering a structurally validated scaffold for engineering selectivity by loop redesign [2].

Figure 1: MMP9–TIMP3 complex (HADDOCK docking structure). Steel blue: MMP9 catalytic domain. Lavender: TIMP3 scaffold. Gold: zinc-binding HExxHxxGxxH motif (HIS120–HIS130) — conserved active site targeted by inhibitor loops. Cyan: TIMP3 AB loop (residues 31–36). Orange: TIMP3 C loop (residues 63–68). These two reactive-site loops insert into the MMP9 active site and serve as the scaffold for loop-expansion selectivity engineering.

Methods: AI-Driven 6-Stage Design Pipeline

The workflow moves from generative structural AI through experimental validation in six tightly coupled stages. (1) A diffusion model generates novel 3D loop backbone geometries on the TIMP3 scaffold across the AB, C, and EF binding loops (6–24 aa expansions) [5], run on an A30 GPU (~20 hr/run). (2) A graph neural network sequences each backbone—sampling 1,000 diverse amino acid sequences per loop/target combination at low temperature (T=0.2) to preserve backbone compatibility while maximizing sequence diversity [6]. The scaffold residues are held fixed; only loop positions are designed. (3) The top 10 sequences per loop are co-folded with each metalloproteinase target using the AlphaFold3 Server; interface confidence metrics (ipTM, pLDDT, Interface PAE) are extracted per complex [4]. (4) A self-normalized T-score filter identifies variants that outperform their own cross-target mean—explicitly selecting for target-preferential binders rather than globally high-confidence structures. Candidates require T>2.0 on ≥2 independent metrics and ipTM ≥ 0.82. (5) All 13 final construct DNA sequences are screened for restriction enzyme recognition sites (BsrGI, BamHI, BsaI) that would interfere with Golden Gate cloning; all 13 passed. (6) Constructs are cloned into a yeast surface display vector, transformed into EBY100 yeast, and screened by flow cytometry. The primary readout is the Positive Median Ratio: the ratio of the APC-A (binding) median to the FITC-A (expression) median within the FITC-positive gate. Dividing by FITC normalizes for surface-display level variation between individual yeast cells while keeping MMP9 and MMP2 values on a shared scale—a higher ratio for MMP9 than MMP2 directly indicates target preference. TIMP3-WT reference lines are drawn at the actual TIMP3-WT Pos Med Ratio per target. Error bars represent 95% confidence intervals (t-distribution). Failed trials (positive control binding efficiency below threshold) are excluded before statistical analysis.

Six-stage workflow illustrated in Figure 2 (center column).

Computational Pipeline

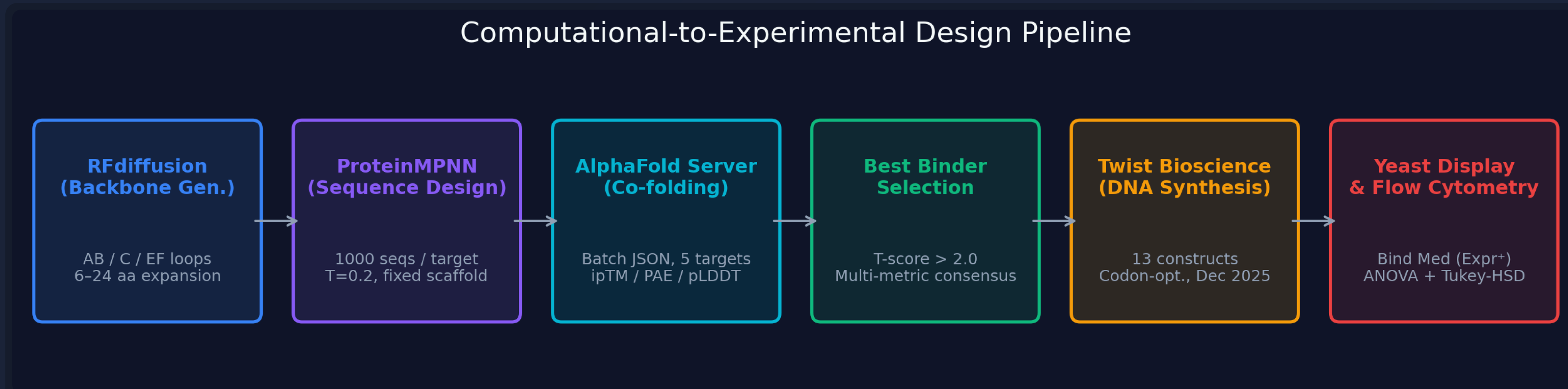

Figure 2: Six-stage generative-AI-to-experiment pipeline. Backbone generation (RFdiffusion) → sequence design (ProteinMPNN) → structural co-folding (AlphaFold3) → T-score candidate selection → DNA synthesis → yeast surface display & flow cytometry. Reduces >1,000 generated sequences to 13 synthesis-ready constructs.

Results

Computational Stage Outcomes

Computational Stage Outcomes: The diffusion backbone generator produced structurally diverse loop geometries in the 6–15 residue range not achievable by conventional mutagenesis, including extended beta-hairpin and helix-turn motifs [5]. Following sequence design and structural co-folding, the self-normalized T-score filter reduced the candidate pool from >1,000 generated sequences to 13 final constructs—a >98% reduction—while enriching for the experimentally confirmed MMP9 binders. Top-ranked MMP9 candidates (C-loop and AB-loop variants) achieved interface confidence scores placing them in the top decile of all evaluated structures, consistent with their subsequent experimental validation.

Experimental Validation

Experimental Validation (MMP9 vs MMP2): Two-group Tukey HSD on the Median Binding Ratio (APC/FITC, FITC+ gate) confirmed statistically significant MMP9 preference over MMP2 for three of four MMP9-targeted constructs (75% PPV). C 12 (loop: ASGPITVNGETIW, C-loop) achieved the highest MMP9 ratio (0.289 vs 0.087 for MMP2), a 3.3× preference ($p=0.009$). C 15 (2.5× preference; $p=0.004$) and AB 6 (loop: TDTFPTANWTGEV, 2.7× preference; $p=0.001$) showed similarly amplified selectivity. AB 1 showed directionally correct binding (0.180 vs 0.080) but did not reach significance ($p=0.277$). Variants predicted as Low binders (C 13, AB 5) showed non-significant results ($p>0.05$), confirming that significant results reflect genuine target discrimination.

Median Binding Ratio Specificity: MMP9 vs MMP2

CONSTRUCT	DESIGN	MMP2	MMP9	TUKEY HSD *P*	VERDICT
C 12	MMP9 Selective	0.087	**0.289**	**0.009**	✓ **MMP9-Selective**
C 15	MMP9 Selective	0.094	**0.232**	**0.004**	✓ **MMP9-Selective**
AB 6	MMP9 Selective	0.078	**0.209**	**0.001**	✓ **MMP9-Selective**
AB 1	MMP9 Predicted	0.080	0.180	0.277	n.s.
AB 2	High Affin M9	—	—	0.300	n.s.
AB 3	High Affinity	—	—	0.341	n.s.
C 13	Low Control	—	—	0.288	n.s.
AB 5	Low Control	—	—	0.529	n.s.
AB 4	ADAM17 Spec	—	—	0.521	n.s.
AB 7	ADAM17 Spec	—	—	0.176	n.s.
C 11	High Affin A17	—	—	0.304	n.s.
C 14	ADAM17 Spec	—	—	0.323	n.s.
TIMP 3	WT Reference	0.107	0.211	0.150	n.s.

Median Binding Ratio = APC-A median / FITC-A median (expression-positive gate). Two-group Tukey HSD per construct (MMP9 vs MMP2 only). n.s. = non-significant. Construct names colored by design group.

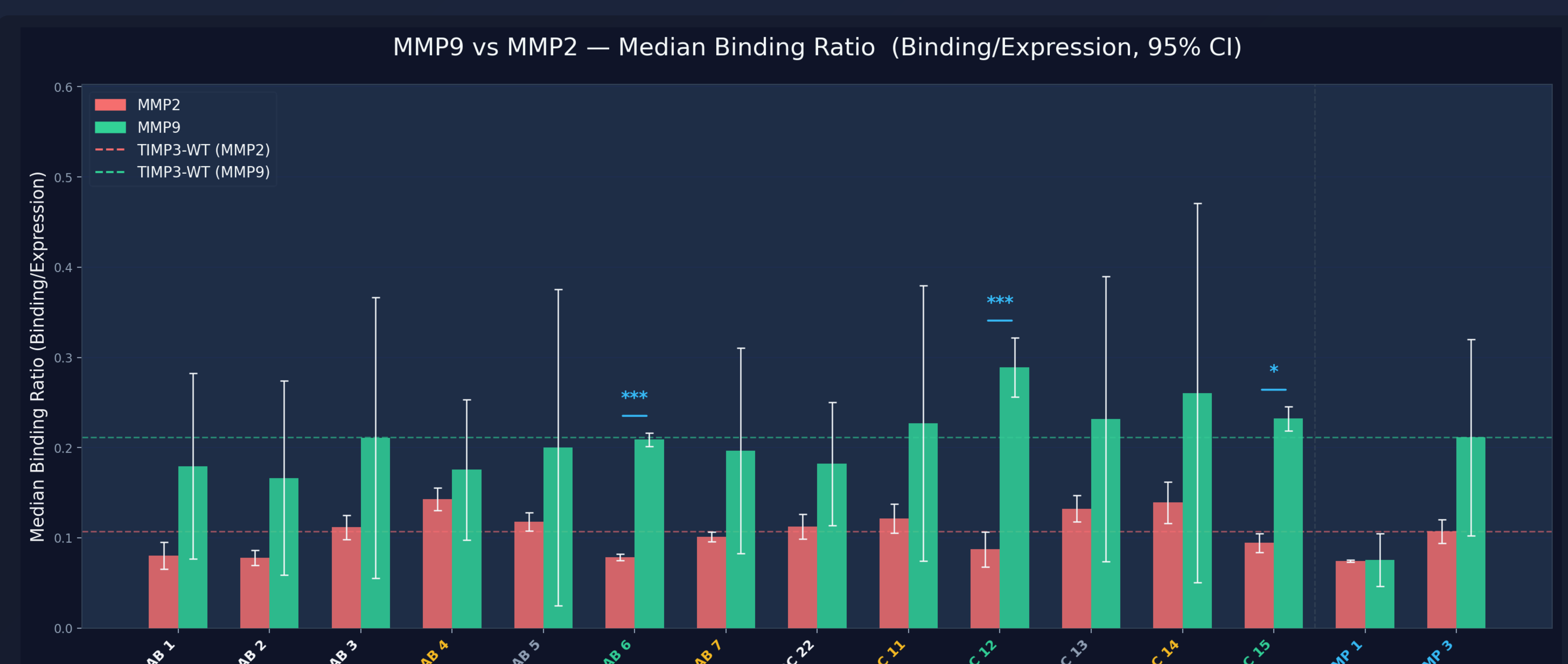

*Figure 3: MMP9 vs MMP2 — Median Binding Ratio (APC-A / FITC-A, expression-positive gate). For each trial, the ratio of the APC-A (binding) median to the FITC-A (expression) median is computed within the FITC+ gate, normalizing for surface-display level variation between cells. Mean ratio ± 95% CI shown. Brackets: Tukey-HSD significant pairs (***$p<0.001$, *$p<0.05$). Dashed lines = TIMP3-WT ratio per target. Construct labels colored by design group (green = ANOVA-confirmed MMP9-selective, amber = ADAM17-targeted, blue = WT reference).*

Conclusions

Pipeline Validation — 3 of 4: Three of four computationally predicted MMP9-selective constructs were experimentally confirmed (75% PPV). AB 1 showed directionally correct binding (MMP9 > MMP2) but high trial-to-trial variability (σ = 43%) prevented statistical confirmation. Every 'Low' control showed non-significant ANOVA—0% false positives. This bidirectional concordance validates the T-score multi-metric filter as a reliable predictor of experimental selectivity, not just overall binding strength. A 75% success rate among specific predictions is strong given the vast landscape of possible protein sequences.

Scaffold Plasticity: Loop insertions of 6–15 residues into the TIMP3 AB and C loops were structurally tolerated (high interface confidence in co-folding) and biologically functional on the yeast surface. Wild-type TIMP3 itself showed significant MMP9 preference ($p=0.024$), and all three engineered MMP9-selective variants substantially amplified this preference.

T-Score vs Z-Score: By normalizing each variant against its own cross-target performance distribution rather than the population, the T-score framework explicitly isolates target-preferential binders. This is the critical distinction: a globally excellent binder (high Z-score) is not necessarily a selective one. The T-score filter directly optimizes for the clinically relevant property.

Pipeline Generalizability: The workflow is target-agnostic. Any protein-protein interaction with available structural data can be addressed using an identical computational pipeline. MMP9 vs MMP2 selectivity serves as the validation case for a generalizable AI-driven precision protein engineering platform.

Future Directions

Kinetic Characterization (Near-term): Flow cytometry measures relative binding in a cell-surface display context. Surface Plasmon Resonance (SPR) will provide absolute dissociation constants (KD), on-rates (kon) and off-rates (koff) for C 12, C 15, and AB 6, enabling direct comparison to wild-type TIMP3 affinity and benchmarking against clinical inhibitor standards.

Reverse Selectivity — MMP2 > MMP9 as Proof of Concept: To rigorously demonstrate that the pipeline can be steered in either direction, we will redesign AB-loop variants specifically targeting MMP2 over MMP9. Since MMP2 plays a homeostatic role and MMP9 drives metastasis, a selective MMP2 binder has less therapeutic utility—but its design and experimental confirmation would constitute the strongest possible proof that the pipeline achieves genuine, controllable specificity rather than an MMP9-intrinsic affinity artifact of the TIMP3 scaffold.

ADAM10/ADAM17 Selectivity (Pending Optimization): ADAM17 data was collected across multiple trial dates and is included in normalized within-target comparisons. ADAM10 trials were excluded due to insufficient positive control signal in all runs. Optimizing the ADAM10 surface protein presentation protocol (titrated antibody concentration, induction time) will enable ADAM17 vs ADAM10 selectivity analysis using the identical ANOVA/T-score framework—extending the specificity pipeline to a clinically relevant Alzheimer's disease (ADAM10) vs inflammatory signaling (ADAM17) distinction.

In Silico Saturation Mutagenesis: A protein language model (trained on evolutionary sequence data from hundreds of millions of natural proteins) will predict the fitness effect of every single-residue substitution in the top MMP9-selective loops. This will identify which positions are critical for selectivity, which tolerate modification, and which substitutions may further improve affinity—dramatically accelerating the next design cycle without additional synthesis.

Clinical & Translational Outlook

MMP9 as an Oncology Target: MMP9 overexpression correlates with tumor invasiveness, lymph node metastasis, and poor prognosis across >15 solid tumor types [3]. The previous failure of pan-MMP inhibitors in clinical trials was not a failure of the therapeutic hypothesis—it was a failure of selectivity. An MMP9-specific TIMP3 variant could inhibit the invasive phenotype while sparing MMP2-dependent basement membrane maintenance in normal tissue, addressing the toxicity problem that terminated earlier drug programs.

A Generalizable AI Protein Engineering Platform: This pipeline's significance extends beyond TIMP3 and MMPs. Any therapeutic context requiring fine specificity between structurally similar targets—kinase isoforms, cytokine receptors, viral surface proteins—is addressable by an identical computational-experimental workflow. The de novo loop engineering approach can generate binding domains for any protein interaction where conventional antibody or small-molecule approaches lack resolution.

Personalized Medicine & CAR-T Cell Analogy: Chimeric Antigen Receptor T-cell (CAR-T) therapies engineer a patient's own immune cells to recognize and destroy tumor cells bearing a specific surface antigen. The fundamental limitation of current CAR-T designs is that shared antigens between tumor and healthy cells cause on-target/off-tumor toxicity—the same selectivity problem faced by MMP inhibitors. A patient's tumor genome can be sequenced to identify neoantigens unique to their cancer. The de novo binder pipeline could then computationally generate binding domains specific to those patient-specific neoantigens and incorporate them into a CAR construct. Combined with AI-driven specificity engineering, this creates a path toward hyperspecific, genome-informed CAR-T therapies with minimal off-tumor toxicity—a convergence of precision genomics and computational protein design.

References

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